GRUS Audit Engine v1.0

Published by: Green Recursive Utility Service LLC (State of Texas)
License: MIT (open source, public reuse permitted)
Specification: GRUS-STD-001-v1.0

This repository contains the open-source GRUS Audit Engine — the automated ten-stage certification pipeline operated by Green Recursive Utility Service LLC for the World Review Framework.

Quick Start

```bash
pip install numpy pyyaml cryptography
python grus_audit_engine/audit_runner.py /path/to/submission --dry-run
```

Pipeline Stages

Stage	Module	Function
1	manifest_parser.py	Validate GRUS_MANIFEST.yaml schema
2	constants_validator.py	Single source of truth + AST hardcoding check
3	doi_resolver.py	Resolve cited DOI; verify priority timestamp
4	dataset_fetcher.py	Download cited public datasets; SHA-256 hash
5	verify_executor.py	Execute submission's verify.py in subprocess
6	residual_comparator.py	Compare verify.py output to manifest residuals
7	closure_tester.py	Cross-domain consistency check
8	falsification_validator.py	Verify pending predictions have falsification specs
9	hasher.py	SHA-256 of manifest, verify.py, constants module
10	certifier.py	Sign certification record with ed25519

Auto-Citation

`cross_reference_engine.py` auto-detects every prior repository element the submission depends on and generates `CITATION_MANIFEST.yaml` with permanent credit attribution to every contributor in the dependency chain.

Compliance

- Executive Order 14303 (Restoring Gold Standard Science, May 23, 2025)
- OSTP Nelson Memo (2022/2026): Public Access Mandate
- Data Integrity Act: deterministic gradient validation, zero-proxy execution

Recursive Auditability

The engine is open-source. Any third party may fork the engine, run it independently against any submission and any prior ledger snapshot, and verify that the engine produces the same result the canonical run produces. Discrepancies are publicly investigable.